quantization of Hamiltonian actions

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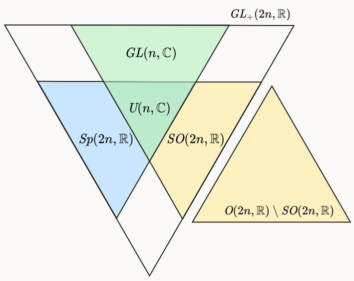
contents

This progression is summarized in this table

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) ***symplectic manifolds***
$J = [i]$	J is the complex structure on the Riemann sphere	$J = \begin{bmatrix} i & & \\ & i & \\ & & \ddots \end{bmatrix}$	We don't need it in general!
$X_H := J\text{grad}(H)$	$X_H := J\text{grad}(H)$	$X_H := J\text{grad}(H)$	$X_H := -\omega^{-1}\mathrm{d}H$ *notice the "$-$" sign!*
$X_H H = 0$	$X_H H = 0$	$X_H H = 0$	$X_H H = 0$
area preserving $\mathcal{L}_{X_H}\mathrm{d}x \wedge \mathrm{d}y = 0$	*area preserving* $\mathcal{L}_{X_H}\lambda_{\text{area}} = 0$	volume preserving, but preserves more...	*"k-dimensional area preserving"* $\mathcal{L}_{X_H}\omega^k = 0$ for every $1 \leq k \leq n$
Converse? True for $\mathbb{R}^2$ but not true for general *open subsets!*	Converse? true for S^2!	Converse for volume preservation is false for $n > 2$, even for linear flows!	Converse is false for $\dim M > 2$! In higher dimensions Hamiltonian flows preserve ω^k for all $1 \leq k \leq n$ so preserving ω becomes a necessary condition. However, it is true that vector fields that preserve the *symplectic 2-form* are Hamiltonian on a simply connected M.
Linear maps that are *generated* by Hamiltonian vector fields are just the area preserving ones $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$	What could be "linear" flows? well isometries $SO(3, \mathbb{R})$ preserve area, so we have one nice Lie group action on the 2-sphere.		symplectic G-actions, Hamiltonian G-actions

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) ***symplectic manifolds***
Is every area preserving flow Hamiltonian? **Yes** But not true for the space $\mathbb{R}^2 \setminus \{0\}$	Is every area preserving flow Hamiltonian? **Yes**	Is every symplectic flow Hamiltonian? **Yes**	Is every symplectic flow Hamiltonian? We have the isomorphism $$H^1(M, \mathbb{R}) \cong H^1(\Omega^\bullet, \mathrm{d}) \cong \frac{\mathrm{Sym}(M, \omega)}{\mathrm{Hamil}(M, \omega)}$$

Some properties, examples of symplectic manifolds follow:

- Symplectic manifold
- Cotangent bundles are symplectic manifolds
- Compatible almost complex structure on a symplectic manifold
- Manifolds with compatible triples

We start to work with coadjoint orbits now...

- Orbits of smooth Lie group actions
- Homogeneous manifold
- Symplectic Lie group action on a symplectic manifold
- Ad^* orbits are symplectic manifolds
- Symplectic homogeneous manifolds
- Hamiltonian Lie group action on a symplectic manifold

Some topics in quantization:

- Geometric quantization of $\mathbb{C}^n$

Now we may start with the orbit method/geometric quantization:

Intuition

We want the following correspondence

symplectic geometry	**representation theory**
Hamiltonian $G \curvearrowright M$	representation $G \curvearrowright \mathcal{H}(M)$
moment map $$\begin{array}{c} M \\ \downarrow{\scriptstyle \mu} \\ \mathfrak{g}^* \\ \coprod_\alpha \mathcal{O}_\alpha \end{array}$$	decomposition into irreps $$\begin{array}{c} \mathcal{H} \\ \downarrow \\ \bigoplus_\alpha E_\alpha \end{array}$$
$$\{f, g\}$$	$$\{A, B\}$$?
orbit $\mathcal{O}_\alpha$	irrep E_α

symplectic geometry	representation theory
symplectic reduction $M//G = \mu^{-1}(0)/G$	$\mathcal{H}(M//G) = \mathcal{H}(M)^G$
Hamiltonian restriction	restricted representation
symplectic induction	induced representation
	Peter-Weyl theorem

[1]

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1. the table in
   - https://chessapig.github.io/talk/G-spaces

   ↩

- Borel-Weil-Bott theorem
- orbit method
    - compact Lie groups
    - Nilpotent Lie groups
- ?...

- On induced representations

references

For symplectic manifolds:

- Jack Lee's book on smooth manifolds
- http://staff.ustc.edu.cn/~wangzuoq/Courses/15S-Symp/SympGeom.html
- Ana Canas da Silver, Lectures on Symplectic Geometry
- [1707.02389] On the universality of potential well dynamics

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For coadjoint orbits, geometric quantization:

- https://chessapig.github.io/talk/G-spaces
- Moment maps and geometric invariant theory
- Matthias Peter - The Orbit Method for Compact Connected Lie Groups
- https://people.math.sc.edu/kellerlv/Symplectic_Geometry_HW1%20(41).pdf
- https://www.cambridge.org/core/journals/canadian-mathematical-bulletin/article/every-symplectic-manifold-is-a-linear-coadjoint-orbit/60EEA4D0397E72FAE554ADEE9EC800B0

motivating Hamiltonian flows

on the plane

We want a vector field that *preserves* a given smooth

$$H : \mathbb{R}^2 \to \mathbb{R}$$

Then

$$\mathrm{grad}(H)$$

does the opposite: its flow goes out of $H^{-1}(c)$ *curves*.

So we should work with the 90 degrees rotation of the gradient, so

$$J\mathrm{grad}(H)$$

where

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is the clockwise 90 degrees rotation matrix.

The flow of this vector field ϕ^t such that

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi^t(p) = J\mathrm{grad}(H)_{\phi^t(p)}$$

which quickly implies

$$\frac{\mathrm{d}}{\mathrm{d}t}H(\phi^t(p)) = \mathrm{d}H(\frac{\mathrm{d}}{\mathrm{d}t}\phi^t(p)) = \langle \mathrm{grad}(H), J\mathrm{grad}(H)\rangle = 0$$

by chain rule!

Hamiltonian vector fields on plane preserve area

$$X_H = \begin{bmatrix} -\frac{\partial H}{\partial y} \\ \frac{\partial H}{\partial x} \end{bmatrix}$$

$$\mathrm{div}(X_H) = -\frac{\partial}{\partial x}\frac{\partial H}{\partial y} + \frac{\partial}{\partial y}\frac{\partial H}{\partial x} = 0$$

And divergence free vector fields preserve area (volume in $\mathbb{R}^n$)

are all area preserving flows Hamiltonian vector fields?

Well,

- $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ is *Hamiltonian* for some smooth H
  - $\iff$

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} -\frac{\partial H}{\partial y} \\ \frac{\partial H}{\partial x} \end{bmatrix}$$

- $\iff$

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \mathrm{d}H = X_2 \mathrm{d}x - X_1 \mathrm{d}y$$

- So we may use facts about 1-forms

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \leftrightarrow \alpha = X_2 \mathrm{d}x - X_1 \mathrm{d}y$$

  - We know **on a simply connected domain** α is closed ($\mathrm{d}\alpha = 0$) $\iff$ α is exact $\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \alpha = \mathrm{d}H$.
  - Now

$$\mathrm{d}\alpha = \underbrace{\left(-\frac{\partial X_1}{\partial x} - \frac{\partial X_2}{\partial y}\right)}_{-\mathrm{div}(X)} \mathrm{d}x \wedge \mathrm{d}y$$

  - Hence, on a **simply connected domain** flow of X is area preserving $\iff$ X is divergence free $\iff$ $\exists H \in \mathcal{C}^\infty(\mathbb{R}^2)$ such that

$$X = J\mathrm{grad}(H)$$

  - **Notice a minus sign pops out in $\mathrm{d}\alpha$ above.**

a area preserving flow that is *not* globally Hamiltonian on the punctured plane

The 1-form

Definition. Angle form on $\mathbb{R}^2 \setminus \{0\}$

The angle form on $\mathbb{R}^2 \setminus \{0\}$ is defined by

$$\frac{-y\mathrm{d}x + x\mathrm{d}y}{x^2 + y^2}$$

using the coordinates (x, y).

is closed but not exact.

So equating this form with $-\alpha$ above we get the vector field

$$X_\theta := \frac{1}{x^2 + y^2} \begin{bmatrix} x \\ y \end{bmatrix}$$

which has zero divergence but does not have a globally defined Hamiltonian.

It is however locally Hamiltonian for the angle function

$$X_\theta = J\mathrm{grad}(\theta)$$

defined on any simply connected subset of $\mathbb{R}^2 \setminus \{0\}$.

Hamiltonian but in different coordinates

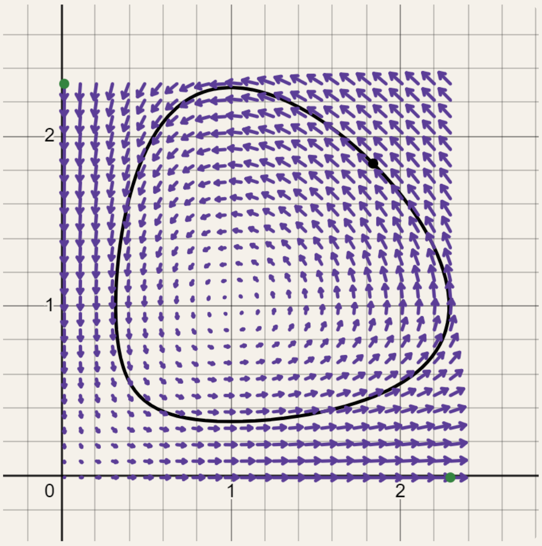
space.f.field.usual Lotka-Volterra

Lotka-Volterra vector field on $\mathbb{R}^2$

Definition. Lotka-Volterra vector field on $\mathbb{R}^2$

$$\begin{bmatrix} \lambda_1 x - \eta_1 xy \\ -\lambda_2 y + \eta_2 xy \end{bmatrix}$$

which looks like



on $\mathbb{R}^2_{>0}$
[1]

for $\lambda_i, \eta_i = 1$

- $$\operatorname{div} \begin{bmatrix} x - xy \\ -y + xy \end{bmatrix} = 1 - y - 1 + x = x - y$$

Intuition

The flow is not area preserving of course, but it has a "*hidden area preserving*" because the flow is periodic on $\mathbb{R}^2_{>0}$.

- However, it preserves a *different* area form on $\mathbb{R}^2_{>0}$

$$\mathfrak{L}_X \left(\frac{\mathrm{d}x \wedge \mathrm{d}y}{xy} \right) = \operatorname{div} \left(\frac{1}{xy} X \right) = 0$$

- Consider the smooth function for $(x, y) \in \mathbb{R}^2_{>0}$

$$x - \log x + y - \log y$$

- The vector field preserves this function

$$\begin{aligned}\partial_{\begin{bmatrix} x-xy \\ -y+xy \end{bmatrix}}(x-\log x+y-\log y) &= (x-xy)\left(1-\frac{1}{x}\right)+(-y+xy)\left(1-\frac{1}{y}\right) \\ &= (x-xy-y+xy)-1+y+1-x \\ &= 0\end{aligned}$$

- However,

$$\begin{aligned}\mathrm{grad}(x-\log x+y-\log y) &= \begin{bmatrix} 1-\frac{1}{x} \\ 1-\frac{1}{y} \end{bmatrix} \\ \implies J\mathrm{grad}(x-\log x+y-\log y) &= \begin{bmatrix} -1+\frac{1}{y} \\ 1-\frac{1}{x} \end{bmatrix} = \frac{1}{xy}\begin{bmatrix} x-xy \\ -y+xy \end{bmatrix}\end{aligned}$$

- This shows the vector field is Hamiltonian vector field with the symplectic form $\frac{1}{xy}\mathrm{d}x \wedge \mathrm{d}y$ on $\mathbb{R}^2_{>0}$.

https://www.desmos.com/calculator/l0gvaeds5s

Lotka-Volter flow

in logarithmic coordinates

- In coordinates

$$\begin{aligned}\mathbb{R}^2_{>0} &\to \mathbb{R}^2_{>0} \\ \begin{pmatrix} x \\ y \end{pmatrix} &\mapsto \begin{pmatrix} \log x \\ \log y \end{pmatrix}\end{aligned}$$

  we have

Now it is a Hamiltonian system.

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1. https://www.desmos.com/calculator/l0gvaeds5s ↩

Todo

- Lotka-Volterra's vector field
- diagonal action on $\mathbb{C}^n$

sidenote: vector fields that are (locally) Hamiltonian as well as (locally) gradient

object	equivalent to Cauchy-Riemann operator	Cauchy-Riemann operator/equivalent gives 0 which makes the object a	if Cauchy-Riemann operator/equivalent gives 0 then a primitive exists *locally*
a complex function $$f = u + iv$$	the Cauchy-Riemann operator $$\frac{\partial}{\partial \bar{z}} f = \frac{1}{2}\Big((u_x - v_y) + i(v$$	Holomorphic function	$\exists g$ $$\frac{\partial g}{\partial \bar{z}} = f$$
a (complex) differential $(1,0)$-form $$f(z)\mathrm{d}z$$	the exterior derivative $$\mathrm{d}(f(z)\mathrm{d}z) = -\frac{\partial f}{\partial \bar{z}}\mathrm{d}z \wedge \mathrm{d}\bar{z}$$	closed differential 1-form/Holomorphic 1-form	$\exists g\mathrm{d}z$ $$\mathrm{d}g = f\mathrm{d}z$$
(real) differential 1-form $$\omega = u\mathrm{d}x - v\mathrm{d}y$$ and its Hodge dual $$\star\omega = v\mathrm{d}x + u\mathrm{d}y$$	$$\mathrm{d}\omega = (u_y + v_x)\mathrm{d}x \wedge \mathrm{d}y$$ $$\mathrm{d}\star\omega = (v_y - u_x)\mathrm{d}x \wedge \mathrm{d}y$$	closed, coclosed (real) differential 1-form	$\exists \alpha, \beta$ (real) 1-forms $$\mathrm{d}\alpha = \omega, \mathrm{d}\beta = \star\omega$$
(real) vector field $$\bar{f} = \begin{bmatrix} u \\ -v \end{bmatrix}$$	$$\mathrm{curl}\bar{f} = -v_x - u_y$$ $$\mathrm{div}\bar{f} = u_x - v_y$$	incompressible (area preserving/symplectic) and a solenoid (curl free) vector field	$\exists H, \phi$ such that $$J\mathrm{grad}(H) = \bar{f} = \mathrm{grad}(\phi)$$

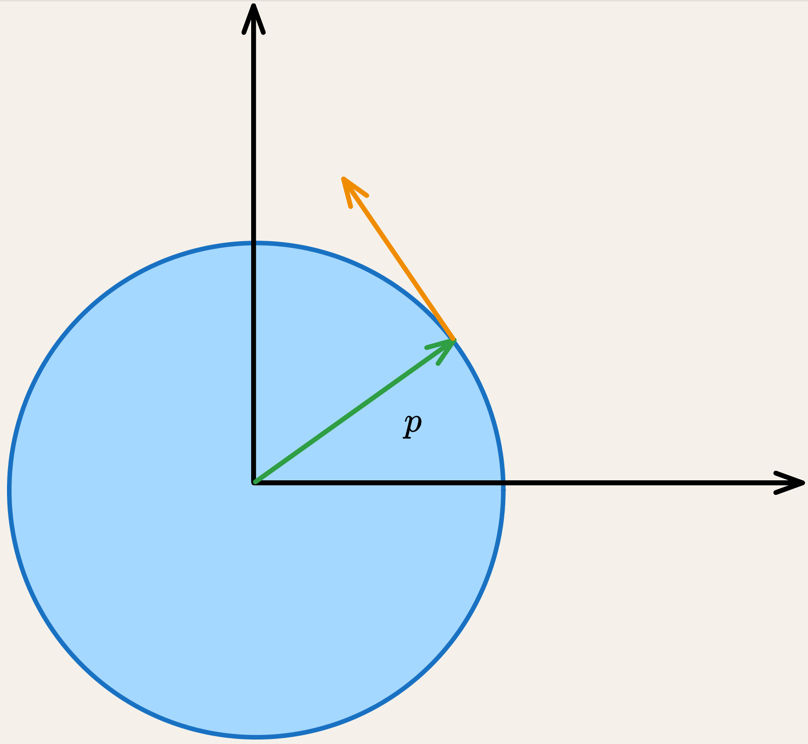
on the 2-sphere

Almost complex structure on S^2 induced from spherical metric and its area form

Definition. Almost complex structure on S^2 using cross product on $\mathbb{R}^3$.

The involution

$$\begin{aligned} J : TS^2 &\to TS^2 \\ X_p &\mapsto p \times X_p \end{aligned}$$



where we've identified $T_pS^2 \leq \mathbb{R}^3$.

Because area spanned by the parallelogram is

$$\|X\| \, \|Y\| \sin(\theta_{X,Y})$$

and *oriented area* is

- supposed to be

$$\begin{aligned} \langle X, Y \rangle &= \|X\| \, \|Y\| \cos(\theta_{X,Y}) \\ &= \|X\| \, \|Y\| \sin\left(\theta_{X,Y} + \frac{\pi}{2}\right) \\ &= \langle JX, Y \rangle \end{aligned}$$

- $$\langle JX, Y \rangle = \langle p \times X, Y \rangle = \langle p, X \times Y \rangle$$

  which, again, *should* be the oriented area of $X, Y \in T_pS^2$
- In local coordinates

$$\langle JX, Y \rangle = \left\langle J \frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right\rangle \mathrm{d}x \wedge \mathrm{d}y = \left\| \frac{\partial}{\partial x} \times \frac{\partial}{\partial y} \right\| \mathrm{d}x \wedge \mathrm{d}y$$

so we conclude

$$g_{\mathsf{S}}(JX, Y) = \lambda_{\mathsf{S}}(X, Y)$$

So same as before we define

Definition. **Hamiltonian vector field of a smooth function on the 2-sphere $H \in \mathcal{C}^\infty(S^2)$ is the vector field**

$$X_H = J\mathrm{grad}(H)$$

- It preserves H because of chain rule, again.
- It preserves area

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space.R.sphere.2.cylindrical

Cylindrical coordinates on S^2

The *cylindrical coordinates*

$$\begin{aligned} S^1 \times (-1,1) &\to S^2 \setminus \{N, S\} \\ (\theta, z) &\mapsto \begin{bmatrix} \sqrt{1-z^2}\cos\theta \\ \sqrt{1-z^2}\sin\theta \\ z \end{bmatrix} \end{aligned}$$

defines a *polar* coordinate chart on S^2 minus the north and south poles.

- The coordinate basis are
  - $$\frac{\partial}{\partial\theta} = \begin{bmatrix} -\sqrt{1-z^2}\sin\theta \\ \sqrt{1-z^2}\cos\theta \\ 0 \end{bmatrix}$$
  - $$\frac{\partial}{\partial z} = \begin{bmatrix} -\frac{1}{\sqrt{1-z^2}} z\cos\theta \\ -\frac{1}{\sqrt{1-z^2}} z\sin\theta \\ 1 \end{bmatrix}$$
- As

  $$\left\langle \frac{\partial}{\partial\theta}, \frac{\partial}{\partial z} \right\rangle = z\sin\theta\cos\theta - z\sin\theta\cos\theta = 0$$

  we have a diagonal **metric**

  $$g_{S^2} = \begin{bmatrix} 1-z^2 & \\ & \frac{1}{1-z^2} \end{bmatrix}$$

  - As

    $$\det \begin{bmatrix} 1-z^2 & \\ & \frac{1}{1-z^2} \end{bmatrix} = 1$$

    we have the **volume (area) form**

    $$\lambda_g = \mathrm{d}\theta \wedge \mathrm{d}z$$

This makes the diffeomorphism from cylinder to sphere minus the poles

$$S^1 \times (-1,1) \to S^2 \setminus \{N, S\}$$

a *volume (area) preserving* diffeomorphism.

- $$\det g_{S^2} = \left\| \frac{\partial}{\partial \theta} \times \frac{\partial}{\partial z} \right\| = 1$$

- The **complex structure**

$$J_p X = \hat{p} \times X$$

and

$$\hat{p} = \frac{\partial}{\partial \theta} \times \frac{\partial}{\partial z}$$

so

$$\begin{aligned} J_p X &= \hat{p} \times X \\ &= \left(\frac{\partial}{\partial \theta} \times \frac{\partial}{\partial z} \right) \times X \\ &= -\left\langle X, \frac{\partial}{\partial z} \right\rangle \frac{\partial}{\partial \theta} + \left\langle X, \frac{\partial}{\partial \theta} \right\rangle \frac{\partial}{\partial z} \end{aligned}$$

- **Gradient vector fields** looks like

$$\begin{aligned} g(\mathrm{grad}(H), -) &= \mathrm{d}H \\ \implies \mathrm{grad}(H)^{\mathsf{T}} \begin{bmatrix} 1 - z^2 & \\ & \frac{1}{1-z^2} \end{bmatrix} &= \left[\frac{\partial H}{\partial \theta}, \frac{\partial H}{\partial z} \right] \\ \implies \mathrm{grad}(H)^{\mathsf{T}} &= \left[\frac{1}{1-z^2} \frac{\partial H}{\partial \theta}, (1 - z^2) \frac{\partial H}{\partial z} \right] \end{aligned}$$

diagonal $T^2 \curvearrowright \mathbb{C}^2$

AKA the dynamics of 2 harmonic oscillators

On the symplectic 4-manifold

$$\left(\mathbb{R}^4, \sum_{i=1}^{2} \mathrm{d}p_i \wedge \mathrm{d}q_i\right) = \left(\mathbb{C}^2, \frac{1}{2i} \sum_{i=1}^{2} \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_i\right)$$

we have the following Hamiltonian vector field and flow

equations	**vector field**	**flow**	**Hamiltonian**
$\dot{p}_1 = -a_1 q_1$ $\dot{q}_1 = a_1 p_1$ $\dot{p}_2 = -a_2 q_2$ $\dot{q}_2 = a_2 p_2$ or in more revealing, matrix form $\begin{bmatrix}\dot{p}_1\\ \dot{q}_1\\ \dot{p}_2\\ \dot{q}_2\end{bmatrix} = \begin{bmatrix} & -a_1 & & \\ a_1 & & & \\ & & & \\ & & a_2 & \end{bmatrix}$ – where the matrix is a block matrix with a scaled $\pi/2$-rotation matrix	$\omega_1 \underbrace{\left(-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1}\right)}_{X_1} +$		$\omega_1 \underbrace{\frac{p_1^2 + q_1^2}{2}}_{H_1} + \omega_2 \underbrace{\frac{p_2^2 + q_2^2}{2}}_{H_2}$
$\dot{z}_1 = i\omega_1 z_1$ $\dot{z}_2 = i\omega_2 z_2$ or $\begin{bmatrix}\dot{z}_1\\ \dot{z}_2\end{bmatrix} = \begin{bmatrix} i\omega_1 & \\ & i\omega_2\end{bmatrix}\begin{bmatrix} z_1\\ z_2\end{bmatrix}$	$\omega_1 \underbrace{i z_1 \frac{\partial}{\partial z_1}}_{X_1} + \omega_2 \underbrace{i z_2 \frac{\partial}{\partial z_2}}_{X_2}$ $= \omega_1 X_1 + \omega_2 X_2$	$\begin{bmatrix} z_1\\ z_2\end{bmatrix} \overset{t}{\mapsto} \begin{bmatrix}\exp(i\omega_1 t) z_1\\ \exp(i\omega_2 t) z_2\end{bmatrix}$	$\omega_1 \underbrace{\frac{z_1 \bar{z}_1}{2}}_{H_1} + \omega_2 \underbrace{\frac{z_2 \bar{z}_2}{2}}_{H_2}$

We may rephrase in terms of the diagonal action on $\mathbb{C}^2$ by $\mathbb{T}^2$

action	generating vector field	moment map
diagonal $\mathbb{T}^2$-action on $\mathbb{C}^2$ *with both frequencies = 1* given by $\mathbb{T}^2 \curvearrowright \mathbb{C}^2$ $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \xmapsto{t_1,t_2} \begin{bmatrix} \exp(it_1)z_1 \\ \exp(it_2)z_2 \end{bmatrix}$	the generators of the $\mathbb{T}^2$-action are the vector fields X_1, X_2 with $[X_1, X_2] = 0$	the H_1, H_2 above with $\{H_1, H_2\} = 0$ give with moment map $(H_1, H_2) : \mathbb{C}^2 \to \mathbb{R}^2$ $(z, z_2) \mapsto \frac{1}{2}(z_1\bar{z}_1, z_2\bar{z}_2)$ of the Hamiltonian $\mathbb{T}^2$-action
from the homomorphism $S^1 \to \mathbb{T}^2 \curvearrowright \mathbb{C}^2$ $t \mapsto l_1 t + l_2 t$ for $l_1, l_2 \in \mathbb{Z}$ we get a S^1-action on $\mathbb{C}^2$ with *integer frequencies*	the generator is $l_1 X_1 + l_2 X_2$	
$\mathbb{R} \to \mathbb{T}^2$	$\omega_1 X_2 + \omega_2 X_2$	
$\mathbb{T}^2 \to \mathbb{T}^2$		

restricting to the invariant 3-sphere

We restrict to the 3-sphere

$$S_h^3 = \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 + p_2^2 + q_2^2 = 2h\} \cong_{\mathsf{Man}} S^3$$

which is invariant under the flow of $\omega_1 X_1 + \omega_2 X_2$ and has orbits of $H_1 = h$. On this sphere, we have the invariant tori

$$T_{h_1,h_2}^2 := \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 = 2h_1, p_2^2 + q_2^2 = 2h_2\}$$

with $\omega_1 h_1 + \omega_2 h_2 = h$.

The projection

$$\begin{aligned} S^3 \setminus \{(0,0,0,1)\} &\to \mathbb{R}^3 \\ (x, y, z, w) &\mapsto (X, Y, Z) := \frac{1}{1-w}(x, y, z) \end{aligned}$$

maps

- the orbit

  $$(p_1, q_1, 0, 0) \mapsto (p_1, q_1, 0)$$

  where $p_1^2 + q_1^2 = 2h_1$, the image is thus a circle in the x-y plane in $\mathbb{R}^3$

- the orbit

$$(0,0,p_2,q_2) \mapsto \left(0,0,\frac{p_2}{1-q_2}\right)$$

  whose image is the z-axis

- $$T^2_{h_1,h_2} \mapsto \left\{(X,Y,Z)\middle| X^2+Y^2=\frac{2h_1}{(1-w)^2}, Z^2=\frac{2h_2-w^2}{(1-w)^2}, w\neq 1\right\}$$

  which should be a torus in $\mathbb{R}^3$

-

restricting to the invariant tori: a torus translation

The subset

$$\begin{aligned} T^2_{a,b} &:= \{(p_1,q_1,p_2,q_2)\in\mathbb{R}^4 | p_1^2+q_1^2=2a, p_2^2+q_2^2=2b\} && \subseteq (\mathbb{R}^2\setminus\{0\})^2 \\ S^3_h &:= \{(p_1,q_1,p_2,q_2)\in\mathbb{R}^4 | p_1^2+q_1^2+p_2^2+q_2^2=2h\} && \subseteq \mathbb{R}^4\setminus\{0\} \end{aligned}$$

which is preserved by the flow of $\omega_1 X_1 + \omega_2 X_2$.

Thus $(\mathbb{R}^2\setminus\{0\})^2$ has a torus fibration

coordinates	X_1	$\omega_1 X_1 + \omega_2 X_2$	flow
On $\mathbb{R}^4$ we have (p_1,q_1,p_2,q_2)	$-q_1\frac{\partial}{\partial p_1}+p_1\frac{\partial}{\partial q_1}$	$\omega_1\underbrace{\left(-q_1\frac{\partial}{\partial p_1}+p_1\frac{\partial}{\partial q_1}\right)}_{X_1}+$	

coordinates	X_1	$\omega_1 X_1 + \omega_2 X_2$	flow
On $(\mathbb{R}^2 \setminus \{0\})^2$ we have the coordinates $r_1^2 = p_1^2 + q_1^2$, θ_1, $r_2 = p_2^2 + q_2^2$, θ_2	$\frac{\partial}{\partial \theta_1}$	$\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}$ which is a torus translation $T^2_{a,b}$	

Flow of a general vector field

$$\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}$$

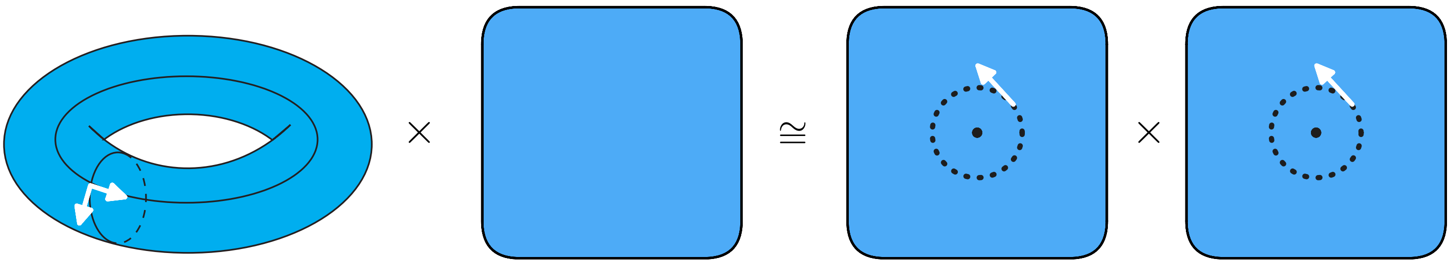
is given by

$$\exp\left(t\left(\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}\right)\right) \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + t \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}$$

whose flows have properties

if	flow is	which gives a	that is coming from
$\omega_1/\omega_2 \in \mathbb{Q}$	periodic	S^1-action on T^2	composition $S^1 \hookrightarrow T^2 \curvearrowright T^2$
$\omega_1/\omega_2 \in \mathbb{R} \setminus \mathbb{Q}$	not periodic, minimal (orbits are dense)	$\mathbb{R}$-action on T^2	composition $\mathbb{R} \hookrightarrow T^2 \curvearrowright T^2$

decomposition into torus translation $\times$ trivial



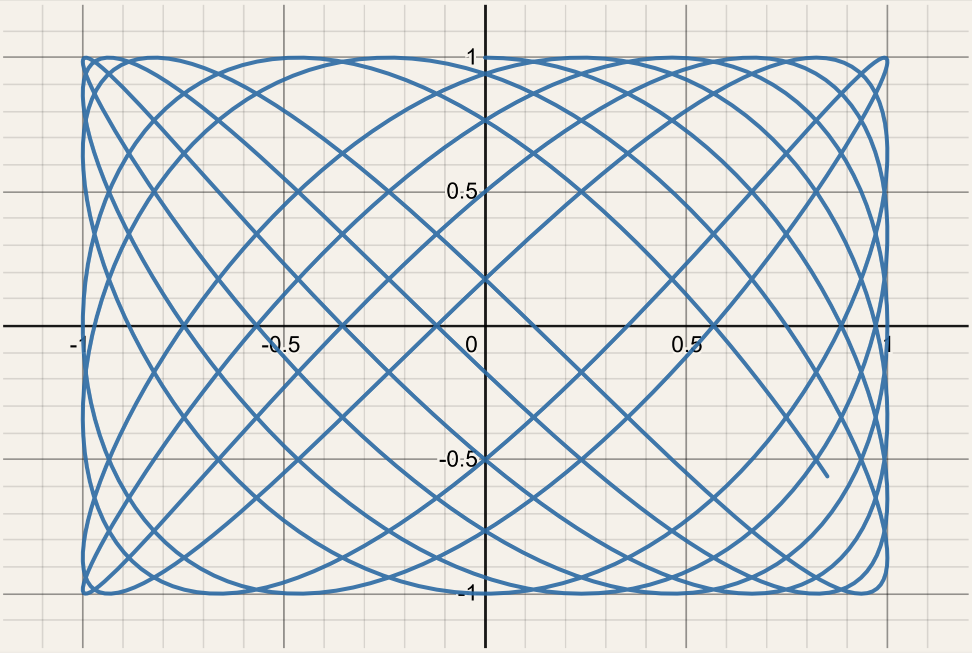
$$\frac{\partial}{\partial \theta_1} \qquad 0 \qquad \longmapsto \qquad -q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1} \qquad 0$$

$$\frac{\partial}{\partial \theta_2} \qquad 0 \qquad \longmapsto \qquad 0 \qquad -q_2 \frac{\partial}{\partial p_2} + p_2 \frac{\partial}{\partial q_2}$$

$$
\begin{aligned}
T^2 \curvearrowright T^2 \times \mathbb{R}^2 &\cong_{T^2} (\mathbb{R}^2 \setminus \{0\})^2 \\
v_1 \mapsto \frac{\partial}{\partial \theta_1} &\times 0 \mapsto X_1 \\
v_2 \mapsto \frac{\partial}{\partial \theta_2} &\times 0 \mapsto X_2
\end{aligned}
$$

projecting onto (q_1, q_2)

$\{(\sin 45t, \cos 65t) | t \in [0,1]\}$



Projecting $(0, q_1, 0, q_2)$ we have

$$(q_1, q_2) \overset{t}{\mapsto} (\sin(\omega_1 t) q_1, \sin(\omega_2 t) q_2)$$

[1]

rational oscillator when $\omega_1, \omega_2 \in \mathbb{Z}$

A $S^1 \cong U(1)$-action (or representation) on $\mathbb{R}^4 = \mathbb{C}^2$

irrational oscillator when $\omega_1/\omega_2 \in \mathbb{R} \setminus \mathbb{Q}$

as a Lax pair

as a Lax pair

We define

$$
\begin{aligned}
L &:= \begin{bmatrix} p & q \\ q & -p \end{bmatrix} \\
M &:= \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}
\end{aligned}
$$

then

$$
\begin{aligned}
[M, L] &= \frac{\omega}{2}\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}\begin{bmatrix}p & q\\q & -p\end{bmatrix} - \begin{bmatrix}p & q\\q & -p\end{bmatrix}\frac{\omega}{2}\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}\\
&= \frac{\omega}{2}\begin{bmatrix}-q & p\\p & q\end{bmatrix} - \frac{\omega}{2}\begin{bmatrix}q & -p\\-p & -q\end{bmatrix}\\
&= \omega\begin{bmatrix}-q & p\\p & q\end{bmatrix}
\end{aligned}
$$

Hence, the oscillator is equivalent to

$$\dot{L} = [M, L]$$

thus giving us conserved quantities

$$\operatorname{tr}(L^2) = \operatorname{tr}(\begin{bmatrix}p^2+q^2 & ?\\? & 0\end{bmatrix}) = p^2+q^2$$

$$
\begin{aligned}
L &:= \begin{bmatrix}p_1 & q_1\\q_1 & -p_1\end{bmatrix} + \begin{bmatrix}p_2 & q_2\\q_2 & -p_2\end{bmatrix}z\\
M &:= \frac{\omega}{2}\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}
\end{aligned}
$$

$$[M, A_1 + A_2 z] = [M, A_1] + [M, A_2]z$$

then

$$\dot{L} = [M, L]$$

should give the equation for isotropic harmonic oscillator.

Now

$$
\begin{aligned}
yI_2 - A_1 - A_2 z &= \begin{bmatrix}y - p_1 - zp_2 & -q_1 - zq_2\\-q_1 - zq_2 & y + p_1 + zp_2\end{bmatrix}\\
\det(yI_2 - A_1 - A_2 z) &= y^2 - (p_1 + zp_2)^2 - (q_1 + zq_2)^2\\
&= y^2 - (p_1^2 + q_1^2) - (p_2^2 + q_2^2)z^2 - 2z(p_1 p_2 + q_1 q_2)
\end{aligned}
$$

which is a conic section over which

$$(y, z) \mapsto \ker(yI_2 - A(z))$$

defines a line bundle.

[2]

---

1. https://www.desmos.com/calculator/irnyqunfsm ↩
2. https://www.desmos.com/calculator/2n2puejwvk ↩

Hamiltonian flows on symplectic manifold

We want a smooth tensor

$$\omega : T^*M \to TM$$

that is, smoothly varying (on $p \in M$) linear maps

$$\omega_p : T_p^*M \to T_p^*M$$

with which we want to define

$$X_H := -\omega^{-1}\mathrm{d}H \iff \mathrm{d}H(-) = -\omega(X_H, -)$$

as the Hamiltonian vector field of H such that

- **flow of X_H must preserve H**

$$0 = X_H H = \mathrm{d}H(X_H) = -\omega(X_H, X_H)$$

  $\iff$ ω is **alternating**, so a 2-form
- and **flow of X_H must preserve ω**

$$0 = \mathcal{L}_{X_H}\omega = \iota_{X_H}\mathrm{d}\omega + \cancelto{0}{\mathrm{d}\underbrace{\iota_{X_H}\omega}_{-\mathrm{d}H}}$$

  which is true when $\mathrm{d}\omega = 0$

Thus

> Definition. **A symplectic manifold (M, ω) is a smooth manifold M with a closed non-degenerate 2-form ω.**

properties of the symplectic form

> **Proposition:** ω^n defines a volume form

Darboux local coordinates on symplectic manifolds

The Darboux Theorem

Our next theorem is one of the most fundamental results in symplectic geometry. It is a nonlinear analogue of the canonical form for a symplectic tensor given in Proposition 22.7. It illustrates the most dramatic difference between symplectic structures and Riemannian metrics: unlike the Riemannian case, there is no local obstruction to a symplectic structure being locally equivalent to the standard flat model.

Theorem 22.13 (Darboux). *Let (M,ω) be a $2n$-dimensional symplectic manifold. For any $p \in M$, there are smooth coordinates $\left(x^1,\dots,x^n,y^1,\dots,y^n\right)$ centered at p in which ω has the coordinate representation*

$$\omega = \sum_{i=1}^{n} dx^i \wedge dy^i. \tag{22.5}$$

We will prove the theorem below. Any coordinates satisfying (22.5) theorem are called ***Darboux coordinates***, ***symplectic coordinates***, or ***canonical coordinates***.

Proof of the Darboux theorem. Let ω_0 denote the given symplectic form on M, and let $p_0 \in M$ be arbitrary. The theorem will be proved if we can find a smooth coordinate chart (U_0,φ) centered at p_0 such that $\varphi^*\omega_1 = \omega_0$, where $\omega_1 = \sum_{i=1}^n dx^i \wedge dy^i$ is the standard symplectic form on $\mathbb{R}^{2n}$. Since this is a local question, by choosing smooth coordinates $\left(x^1,\dots,x^n,y^1,\dots,y^n\right)$ in a neighborhood of p_0, we may replace M with an open ball $U \subseteq \mathbb{R}^{2n}$. Proposition 22.7 shows that we can arrange by a linear change of coordinates that $\omega_0|_{p_0} = \omega_1|_{p_0}$.

Let $\eta = \omega_1 - \omega_0$. Because η is closed, the Poincaré lemma (Theorem 17.14) shows that we can find a smooth 1-form α on U such that $d\alpha = -\eta$. By subtracting a constant-coefficient (and thus closed) 1-form from α, we may assume without loss of generality that $\alpha_{p_0} = 0$.

For each $t \in \mathbb{R}$, define a closed 2-form ω_t on U by

$$\omega_t = \omega_0 + t\eta = (1-t)\omega_0 + t\omega_1.$$

Let J be a bounded open interval containing $[0,1]$. Because $\omega_t|_{p_0} = \omega_0|_{p_0}$ is nondegenerate for all t, a simple compactness argument shows that there is some neighborhood $U_1 \subseteq U$ of p_0 such that ω_t is nondegenerate on U_1 for all $t \in \bar{J}$. Because of this nondegeneracy, the smooth bundle homomorphism $\widehat{\omega}_t\colon TU_1 \to T^*U_1$ defined by $\widehat{\omega}_t(X) = X \lrcorner\, \omega_t$ is an isomorphism for each $t \in \bar{J}$.

Define a smooth time-dependent vector field $V\colon J \times U_1 \to TU_1$ by $V_t = \widehat{\omega}_t^{-1}\alpha$, or equivalently

$$V_t \lrcorner\, \omega_t = \alpha.$$

Our assumption that $\alpha_{p_0} = 0$ implies that $V_t|_{p_0} = 0$ for each t. If $\theta\colon \mathcal{E} \to U_1$ denotes the time-dependent flow of V, it follows that $\theta(t, 0, p_0) = p_0$ for all $t \in J$, so $J \times \{0\} \times \{p_0\} \subseteq \mathcal{E}$. Because $\mathcal{E}$ is open in $J \times J \times M$ and $[0,1]$ is compact, there is some neighborhood U_0 of p_0 such that $[0,1] \times \{0\} \times U_0 \subseteq \mathcal{E}$.

For each $t_1 \in [0,1]$, it follows from Proposition 22.15 that

$$\begin{aligned}
\frac{d}{dt}\bigg|_{t=t_1} (\theta_{t,0}^*\omega_t) &= \theta_{t_1,0}^*\left(\mathcal{L}_{V_{t_1}}\omega_{t_1} + \frac{d}{dt}\bigg|_{t=t_1}\omega_t\right)\\
&= \theta_{t_1,0}^*\big(V_{t_1} \lrcorner\, d\omega_{t_1} + d(V_{t_1} \lrcorner\, \omega_{t_1}) + \eta\big)\\
&= \theta_{t_1,0}^*(0 + d\alpha + \eta) = 0.
\end{aligned}$$

Therefore, $\theta_{t,0}^*\omega_t = \theta_{0,0}^*\omega_0 = \omega_0$ for all t. In particular, $\theta_{1,0}^*\omega_1 = \omega_0$. It follows from Theorem 9.48(c) that $\theta_{1,0}$ is a diffeomorphism onto its image, so it is a coordinate map. Because $\theta_{1,0}(p_0) = p_0 = 0$, these coordinate are centered at p_0. $\square$

Moser's theorem on volumes

Theorem 2. (Moser) Assume M is a compact, connected and oriented manifold of dimension m without boundary. If α and β are two volume forms such that their total volumes agree, i.e.,

$$\int_M \alpha = \int_M \beta, \tag{1.40}$$

then there is a diffeomorphism u of M satisfying $u^*\beta = \alpha$.

Consequently the total volume is the only invariant of volume-preserving diffeomorphisms.

Proof. We proceed as in the proof of Darboux's theorem and deform the volume form α into β defining

$$\alpha_t = (1-t)\alpha + t\beta, \quad 0 \le t \le 1.$$

These forms α_t are volume forms since locally α and β are represented by $\alpha = a(x)dx_1 \wedge \ldots \wedge dx_m$ and $\beta = b(x)dx_1 \wedge \ldots \wedge dx_m$ with nonvanishing smooth functions a and b, which, by assumption (1.40), must have the same sign. We shall construct a family φ^t of diffeomorphisms satisfying

$$\left(\varphi^t\right)^*\alpha_t = \alpha, \quad \varphi^0 = id \tag{1.41}$$

for $0 \le t \le 1$, so that the diffeomorphism $u = \varphi^1$ will solve our problem. Since M is compact, connected and oriented we conclude from $\int_M (\beta - \alpha) = 0$ that

$$\beta - \alpha = d\gamma \tag{1.42}$$

for some $(m-1)$-form γ on M. This is a special case of the de Rham theorem. Since α_t is a volume form we find a unique time-dependent vector field X_t on M solving the linear equation

$$i_{X_t}\alpha_t = -\gamma$$

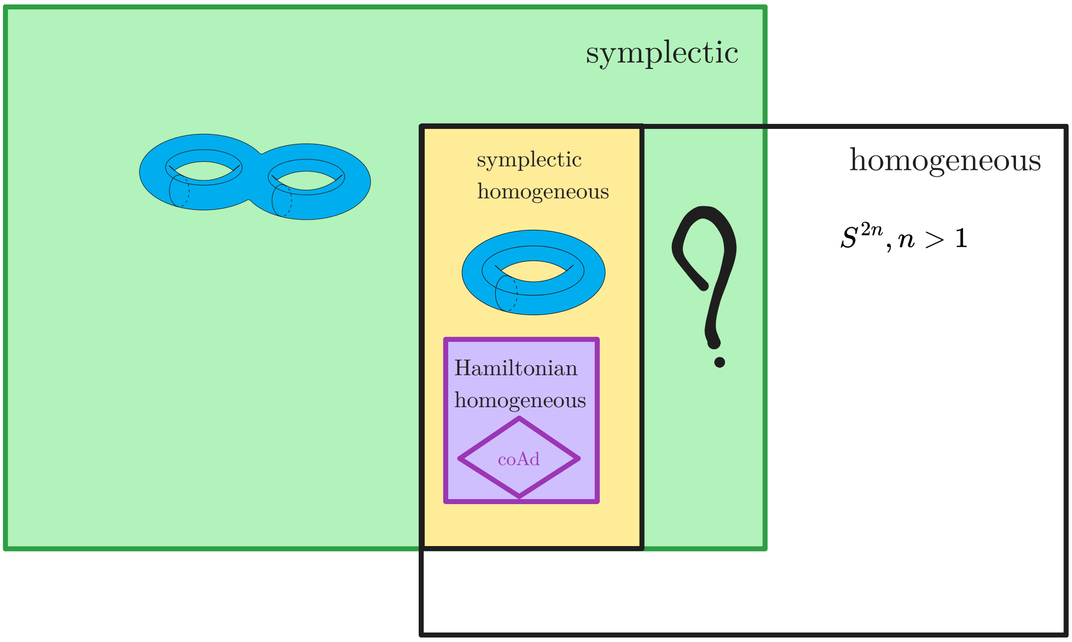
for $0 \leq t \leq 1$. Denote by φ^t the flow of this vector field X_t satisfying $\varphi^0 = id$. Since M is compact it exists for all t. Since $d\alpha_t = 0$ for volume forms we find, again using Cartan's formula,

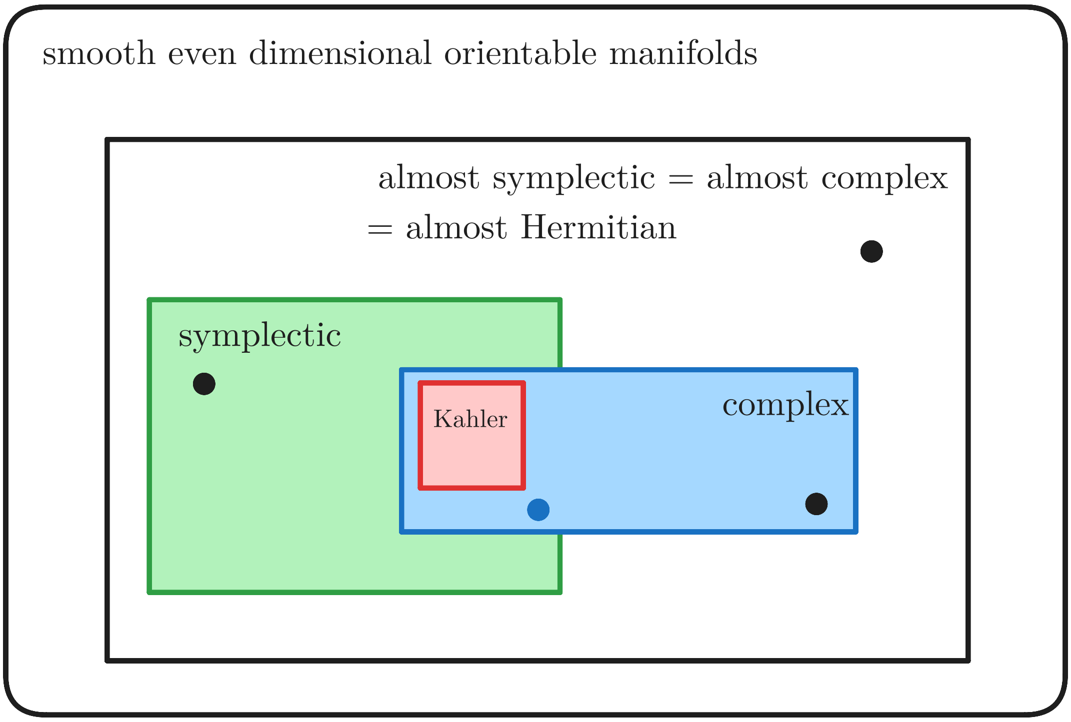
$$\begin{aligned} \tfrac{d}{dt}\left(\varphi^t\right)^*\alpha_t &= \left(\varphi^t\right)^*\left(L_{X_t}\alpha_t + \tfrac{d}{dt}\alpha_t\right) \\ &= \left(\varphi^t\right)^*\left(d(i_{X_t}\alpha_t) + \beta - \alpha\right), \end{aligned}$$

which vanishes since $d(i_{X_t}\alpha_t) + \beta - \alpha = d(i_{X_t}\alpha_t + \gamma) = 0$ by our choice of the vector field X_t. Therefore (1.41) holds and the proof is finished. ■

Symplectic manifold

Definition. **A symplectic manifold (M, ω) is a smooth manifold M with a closed non-degenerate 2-form ω.**





Compatible almost complex structure on a symplectic manifold

Definition. **An almost complex structure J is compatible with a symplectic structure ω if**

$$\omega(-, J-)$$

is a Riemannian metric.

Corollary 12.7. *Any symplectic manifold has compatible almost complex structures.*

– How different can compatible almost complex structures be?

Proposition 12.8. *Let (M,ω) be a symplectic manifold, and J_0, J_1 two almost complex structures compatible with ω. Then there is a smooth family $J_t, 0 \leq t \leq 1$, of compatible almost complex structures joining J_0 to J_1.*

Proof. By compatibility, we get

$$\left.\begin{array}{l} \omega, J_0 \rightsquigarrow g_0(\cdot,\cdot) = \omega(\cdot, J_0\cdot) \\ \omega, J_1 \rightsquigarrow g_1(\cdot,\cdot) = \omega(\cdot, J_1\cdot) \end{array}\right\} \quad \text{two riemannian metrics on } M \ .$$

Their convex combinations

$$g_t(\cdot,\cdot) = (1-t)g_0(\cdot,\cdot) + tg_1(\cdot,\cdot) \ , \qquad 0 \leq t \leq 1 \ ,$$

form a smooth family of riemannian metrics. Apply the polar decomposition to (ω, g_t) to obtain a smooth family of J_t's joining J_0 to J_1. □

Corollary 12.9. *The set of all compatible almost complex structures on a symplectic manifold is path-connected.*

Manifolds with compatible triples

Definition. Smooth manifold with compatible triples

A smooth manifold with a compatible triple

$$(M, \omega, J, g)$$

consists of

- almost symplectic structure ω
- almost complex structure J
- a Riemannian metric g

which satisfy

$$J = g^{-1}\omega^{-1}$$

This gives

- two volume forms λ_g and $\omega^{\dim M/2}$, which *may* differ but of course there exists a never vanishing $s \in \mathcal{C}^\infty(M)$ such that

$$s\lambda_g = \omega^{\dim M/2}$$

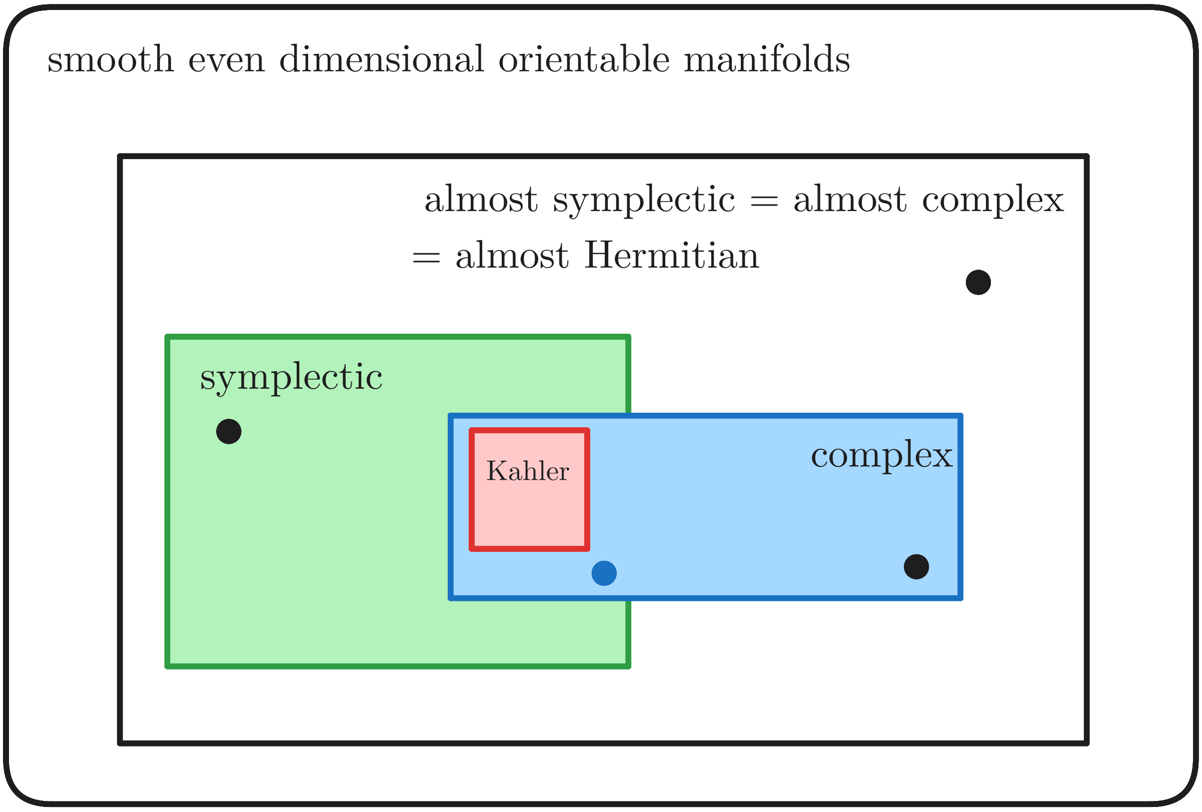
- J-rotation of gradient vector fields **match** with ω-Hamiltonian vector fields

$$\begin{aligned} & g(\mathrm{grad}(f), -) = \mathrm{d}f(-) = -\omega(X_f, -) \\ \implies & g(\mathrm{grad}(f), -) = -g(JX_f, -) \\ \implies & X_f = J\mathrm{grad}(f) \end{aligned}$$

Proposition:

given (ω almost symplectic)	with the condition	we have the third structure
ω, J	$\omega(-, J-)$ is a Riemannian metric	$g := \omega(-, J-)$ is **Riemannian**
ω, g	none	J using polar decomposition
g, J	J is orthogonal, as in $$g(J-, J-) = g(-, -)$$ $$\iff g(J-, -) = -g(-, J-)$$ **(almost Hermitian)**	$\omega := g(J-, -)$ is **almost symplectic**

[1]



1. S Ramanan - Global Calculus-American Mathematical Society (2005), p.193 ↩

Orbits of smooth Lie group actions

Let $\theta : G \curvearrowright M$ be a smooth left Lie group action on a smooth manifold M and $p \in M$. Then the orbit map

$$\begin{aligned}\theta_{(p)} : G &\to M \\ g &\mapsto gp\end{aligned}$$

is smooth and has constant rank, so the isotopy group $G_p = \theta_{(p)}^{-1}(p)$ is a properly embedded Lie subgroup of G. Thus

$$\theta_{(p)} : \frac{G}{G_p} \to M$$

is an injective immersion making the image, that is, the orbit $G\{p\}$ is an immersed submanifold

$$\iota : (G\{p\}, \theta_{(p)*}\mathfrak{T}_{G/G_p}, \theta_{(p)*}\boldsymbol{\mathcal{A}}_{G/G_p}) \hookrightarrow M$$

which is embedded if the action is proper.

Proposition 7.26 (Properties of the Orbit Map). *Suppose θ is a smooth left action of a Lie group G on a smooth manifold M. For each $p \in M$, the orbit map $\theta^{(p)}: G \to M$ is smooth and has constant rank, so the isotropy group $G_p = \left(\theta^{(p)}\right)^{-1}(p)$ is a properly embedded Lie subgroup of G. If $G_p = \{e\}$, then $\theta^{(p)}$ is an injective smooth immersion, so the orbit $G \cdot p$ is an immersed submanifold of M.*

21-17. Suppose a Lie group acts smoothly (but not necessarily properly or freely) on a smooth manifold M. Show that each orbit is an immersed submanifold of M, which is embedded if the action is proper.

Homogeneous manifold

Definition. Homogeneous manifold

A **transitive** smooth Lie group action on a smooth $\mathbb{R}$-manifold is called a homogeneous G-manifold.

$$\frac{G}{G_p} \cong M$$

[1]

Is every smooth manifold homogeneous?

- Given any two points in a connected manifold, there exists a diffeomorphism taking one to the other. [1]
- However, not all smooth manifolds are homogeneous G-spaces for (finite dim) Lie groups G, there are necessary conditions [2]

(Mostow, 2005) M is a compact homogeneous G-space $\implies \chi(M) \geq 0$ [3]

---

1. differential geometry - Proving that given any two points in a connected manifold, there exists a diffeomorphism taking one to the other - Mathematics Stack Exchange ↩
2. dg.differential geometry - Example of a manifold which is not a homogeneous space of any Lie group - MathOverflow ↩
3. A Structure Theorem for Homogeneous Spaces | Geometriae Dedicata ↩

- Symmetric spaces
- Model geometry

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1. mathi.uni-heidelberg.de/~lee/MenelaosSS16.pdf ↩

Symplectic Lie group action on a symplectic manifold

Definition. Symplectic Lie group action on a symplectic manifold

A smooth Lie group action

$$\Phi : G \curvearrowright M$$

on a symplectic manifold is a **symplectic action** if G acts by symplectomorphisms

$$\Phi : G \to \mathrm{Sym}(M, \omega)$$

- For $\dim M = 2$, a symplectic Lie group action is an *area preserving* Lie group action.

Ad^* orbits are symplectic manifolds

Definition. Symplectic structure on coadjoint orbits of Lie groups

Let

$$\Omega \subset \mathfrak{g}^*$$

be a coadjoint orbit $\mathrm{Ad}^* : G \curvearrowright \mathfrak{g}^*$ of a Lie group G. Let $F \in \Omega$.

- By definition Ω is a transitive (left) G-space with stabilizer stabilizer of F being $\mathrm{Ad}^*_{G,F}$ (a closed subgroup).
- Thus the surjective smooth map

$$\begin{aligned} G &\to \Omega \\ g &\mapsto \mathrm{Ad}^*_g(F) \end{aligned}$$

  with descends to bijection

$$\frac{G}{\mathrm{Ad}^*_{G,F}} \to \Omega$$

  giving a smooth structure on the orbit Ω making it a *smooth immersed manifold* in $\mathfrak{g}^*$.
  We construct a anti-symmetric bilinear map

$$\begin{aligned} \omega_F : \frac{\mathfrak{g}}{\mathrm{Lie}(\mathrm{Ad}^*_{G,F})} \times \frac{\mathfrak{g}}{\mathrm{Lie}(\mathrm{Ad}^*_{G,F})} &\to \mathbb{R} \\ (X, Y) &\mapsto F([X, Y]) \end{aligned}$$

Proposition:

- The bilinear map

$$\begin{aligned} \omega_F : \frac{\mathfrak{g}}{\mathrm{Lie}(\mathrm{stab}_G(F))} \times \frac{\mathfrak{g}}{\mathrm{Lie}(\mathrm{stab}_G(F))} &\to \mathbb{R} \\ (X, Y) &\mapsto F([X, Y]) \end{aligned}$$

  defines a Ad^*-invariant **symplectic form** on the orbit $\Omega \subset \mathfrak{g}$.
- The coadjoint action $G \curvearrowright \Omega$ is **Hamiltonian** with moment map

$$\iota : \Omega \hookrightarrow \mathfrak{g}^*$$

  Thus

$$\mathrm{Ad}^* : G \curvearrowright (\Omega, \omega) \subset \mathfrak{g}^*$$

  is a **homogeneous Hamiltonian G-manifold**.

[1]

[2]

Lie group	coadjoint orbits
$SO(3,\mathbb{R}), SU(2,\mathbb{C})$	$\{0\}, S^2$
compact	Kahler [3]
nilpotent	?

---

1. page 48 of the pdf
    - Matthias Peter - The Orbit Method for Compact Connected Lie Groups

    ↩
2. studenttheses.uu.nl/bitstream/handle/20.500.12932/7207/Maes%2CJMA2011.pdf?sequence=1&isAllowed=y#page=40.00
    ↩
3. prop 3.1.21 of the pdf
    - Matthias Peter - The Orbit Method for Compact Connected Lie Groups

    ↩

Symplectic homogeneous manifolds

Definition. **A symplectic homogeneous manifold is a symplectic manifold with a transitive symplectic Lie group action.**

[1]

- Ad^* orbits are symplectic manifolds

classification of homogeneous symplectic manifolds

Let X, G be simply connected and

$$G \curvearrowright (X, \omega)$$

by a homogeneous symplectic G-manifold. Thus it is a **almost Hamiltonian G-action**.

Now define

$$\bar{\mathfrak{g}} := \left\{(h, Z) \in \mathcal{C}^\infty(X) \times \mathfrak{g} \middle| Z = -\omega^{-1}\mathrm{d}h\right\}$$

with bracket

$$[(h_1, Z_1), (h_2, Z_2)] = [\{h_1, h_2\}, [Z_1, Z_2]]$$

then we have the exact sequence

$$\begin{array}{ccccc} 0 \to \mathbb{R} \to \bar{\mathfrak{g}} & \to & \mathfrak{g} \to & 0 \\ (h, Z) & \mapsto & Z & \end{array}$$

The corresponding Lie group $\overline{G}$ we have

$$\overline{G} \to G \curvearrowright X$$

- Now define

$$\begin{aligned} \Phi : X &\to \bar{\mathfrak{g}}^* \\ x &\mapsto \Phi_x : \Phi_x(h, Z) = h(x) \end{aligned}$$

  - Then

$$\begin{gathered} \omega(Z, -) = -\mathrm{d}h \\ \cdots \end{gathered}$$

    so this Φ is indeed a moment map.

  - By construction ...

[2]

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1. Symplectic homogeneous space - Encyclopedia of Mathematics ↩
2. https://people.math.sc.edu/kellerlv/Symplectic_Geometry_HW1%20(41).pdf ↩

Hamiltonian Lie group action on a symplectic manifold

For $\mathbb{R}$,

$$\{\text{symplectic}\ \mathbb{R} \curvearrowright (M,\omega)\} \underset{\exp(t\,-)}{\overset{\frac{d}{dt}}{\rightleftharpoons}} \{\text{complete SymVec}(M,\omega)\}$$

The action is *"Hamiltonian"* if there exists a $H \in \mathcal{C}^\infty(M)$ that is, a smooth map

$$H : M \to \mathbb{R}$$

such that its Hamiltonian vector field generates the $\mathbb{R}$ action

$$\omega^{-1}\mathrm{d}H = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=1} \Phi^t$$

Given a symplectic action

Definition. Symplectic Lie group action on a symplectic manifold

A smooth Lie group action

$$\Phi : G \curvearrowright M$$

on a symplectic manifold is a **symplectic action** if G acts by symplectomorphisms

$$\Phi : G \to \mathrm{Sym}(M,\omega)$$

we have a Lie algebra homomorphism

$$\mathrm{d}\Phi : \mathfrak{g} \to \mathrm{SymVec}(M,\omega)$$

If we demand the image is inside Hamiltonian vector fields

$$\mathrm{d}\Phi : \mathfrak{g} \to \mathrm{HamVec}(M,\omega)$$

then we have

Definition. Hamiltonian Lie group acition on a symplectic manifold

A

Definition. Symplectic Lie group action on a symplectic manifold

A smooth Lie group action

$$\Phi : G \curvearrowright M$$

on a symplectic manifold is a **symplectic action** if G acts by symplectomorphisms

$$\Phi : G \to \mathrm{Sym}(M,\omega)$$

is a **Hamiltonian action** if there exists a smooth map

$$\begin{aligned}\mu : M &\to \mathfrak{g}^* \\ p &\mapsto \mu^{(-)}(p)\end{aligned}$$

such that

- for each $v \in \mathfrak{g}$, the component of μ along v

  $$\mu^v(-) \in \mathcal{C}^\infty(M)$$

  is the function whose Hamiltonian vector field that generates the action by v

  $$\omega^{-1}\mathrm{d}\mu^v = \mathrm{d}_i\Phi(v) \iff \mathrm{d}\mu^v = \omega(\mathrm{d}_i\Phi(v), -)$$

- μ is equivariant with the G action on M and coadjoint action on $\mathfrak{g}^*$

  $$\mu \circ \Phi^g = \mathrm{Ad}_g^* \circ \mu$$

For a basis (v_i) of $\mathfrak{g}$, we have $\dim G$ functions

$$\mu^{v_i} : M \to \mathbb{R}$$

whose Hamiltonian vector field $=$ vector field that generates v_i-actions

$$\omega^{-1}\mathrm{d}\mu^{v_i} = \mathrm{d}\Phi(v)$$

such that the map $\mu : M \to \mathfrak{g}^*$ is equivariant.

So given the moment map we have the G-action

$$\exp t(\omega^{-1}\mathrm{d}\mu^{v_i})$$

Definition. **A symplectic Hamiltonian G-manifold is a** Hamiltonian G-action

$$\Phi : G \underbrace{\curvearrowright}_{\mu} (M, \omega)$$

with moment map μ.

symmetry

On a

Definition. **A symplectic Hamiltonian G-manifold is a** Hamiltonian G-action

$$\Phi : G \underbrace{\curvearrowright}_{\mu} (M, \omega)$$

with moment map μ.

a function

$$f \in \mathcal{C}^\infty(M)$$

is G-invariant $\iff$ μ is a constant on the flow of the Hamiltonian vector field **of f.**

$$L_{X_f}\mu^v = \mathrm{d}\mu^v(X_f) = -\omega(\mathrm{d}\Phi(v), X_f) = -L_{\mathrm{d}\Phi(v)}f$$

[1]

- We have a correspondence between **G-invariant functions f**

$$\Longleftrightarrow$$

$\mathbb{R}$-*flows* $\exp tX_f$ on M such that μ is constant on the flow

$$\Longleftrightarrow$$

$\mathbb{R}$-*flows* $\exp tX_f$ on M that commute with the action of $\Phi : G \curvearrowright M$

$$0 = \omega(\mathrm{d}\Phi(v), X_f) = -[\mathrm{d}\Phi(v), X_f]$$

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1. Ana Cannas da Silva - Lectures on Symplectic Geometry -Springer (2009).pdf > page=158 ↩
2. Lec08.pdf (ustc.edu.cn) ↩

Geometric quantization of $\mathbb{C}^n$

We have $\mathbb{C}^n$ with coordinates (p_j, x_j) and

$$\omega = \sum_j \mathrm{d}p_j \wedge \mathrm{d}x_j = \mathrm{d}\left(\underbrace{\sum_j p_j \mathrm{d}x_j}_{\theta}\right)$$

Definition. Pre-quantization

We define **pre-quantization** to be

$$\begin{aligned} Q_{\text{pre}} : \mathcal{C}^\infty(\mathbb{C}^n) &\to \mathsf{EndVec}(L^2(\mathbb{R}^{2n})) \\ f &\mapsto i\hbar\underbrace{\left(X_f - \frac{i}{\hbar}\theta(X_f)\mathsf{Id}\right)}_{\nabla_{X_f}} + f\mathsf{Id} \\ &= i\hbar X_f + (f + \theta(X_f))\mathsf{Id} \end{aligned}$$

then $1 \mapsto i\hbar(0-0) + 1 = \mathsf{Id}$.

Proposition:

1. $$[\nabla_X, \nabla_Y] = \nabla_{[X,Y]} - \frac{i}{\hbar}\omega(X,Y)$$
2. $$[Q_{\text{pre}}(f), Q_{\text{pre}}(g)] = i\hbar Q_{\text{pre}}(\{f,g\})$$

Quantization

For a symplectic potential θ, define

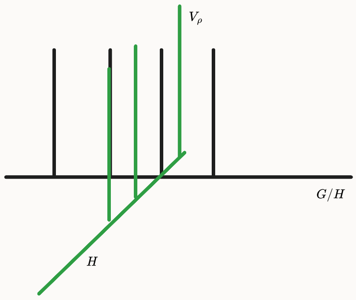
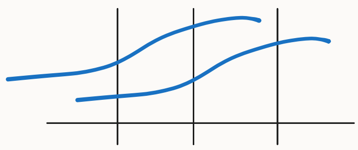
$$\begin{aligned} H_{\mathbf{x}} &:= \left\{\psi \in \mathcal{C}^\infty(\mathbb{C}^n) \middle| \forall j, \nabla_{p_j}\psi = 0\right\} \\ H_{\mathbf{p}} &:= \left\{\psi \in \mathcal{C}^\infty(\mathbb{C}^n) \middle| \forall j,\ \nabla_{x_j}\psi = 0\right\} \\ H_\alpha &:= \left\{\psi \in \mathcal{C}^\infty(\mathbb{C}^n) \middle| \forall j,\ \nabla_{\frac{1}{2}(x_j - \frac{i}{\alpha}p_j)}\psi = 0\right\} \end{aligned}$$

Proposition:

1. $$H_{\mathbf{x}} = \{\psi(\mathbf{x},\mathbf{p}) = \phi(\mathbf{x}) | \phi \in \mathcal{C}^\infty(\mathbb{R}^n)\}$$
2. $$H_{\mathbf{p}} = \left\{\psi(\mathbf{x},\mathbf{p}) = e^{i\mathbf{x}\cdot\mathbf{p}/\hbar}\phi(\mathbf{p}) \middle| \phi \in \mathcal{C}^\infty(\mathbb{R}^n)\right\}$$
3. $$H_\alpha = \left\{\psi(\mathbf{x},\mathbf{p})F(\mathbf{x} - i\alpha\mathbf{p})e^{-\alpha|\mathbf{p}|^2/2\hbar} \middle| F \in \mathscr{O}(\mathbb{C}^n)\right\}$$

$$H^2_{\mathbf{x}} = \left\{\psi(\mathbf{x},\mathbf{p}) = \phi(\mathbf{x}) \middle| \phi \in L^2(\mathbb{R}^n)\right\}$$

Ind reps

	finite case	"H-sections"	sections on the fibred product bundle
		$\begin{array}{c} G \times V_\rho \\ \downarrow \\ G \end{array}$ 	$\begin{array}{ccc} G \curvearrowright & G \times_H V_\rho \\ & \downarrow \\ G \curvearrowright & G/H \end{array}$
Ind_H^G	$V_\pi := \bigoplus_{G/H} V_\rho$	$= \{f : G \to V_\rho \vert f(gh) = \rho($	$\Gamma(G/H, G \times_H V_\rho)$ which are L^2 "a easier version of direct integrals"
G-action	$\pi(gg_i)(v, i) = (\rho(h)v, j)$	$(\pi(g)f)(g) = f(g^{-1}g)$	$(gs)(g_1 H) = gs(g^{-1}g_1 H)$ 
inner product on the bundle			$\langle [g, v], [g, w] \rangle = \langle v, w \rangle_{V_\rho}$ (which is well-defined)
Let ν be a G-invariant measure on G/H (not available in general)		$\langle f_1, f_2 \rangle := \int_{G/H} \langle f_1(g), f_2($	$\langle s_1, s_2 \rangle = \int_{G/H} \langle s_1(gH), s_2$

Proposition:

$$\Gamma(G/H, G \times_H V_\rho) = \Gamma_H(G, G \times V_\rho)$$

$$f \mapsto \left(gH \mapsto [(g, f(h))]\right)$$

- and

$$f_s \leftarrow\!\mid s$$

where f_s is defined to be the element

$$s(gH) = [(g, f_s(g))]$$

> **Proposition:** The induced representation is unitary.

$$\begin{aligned}\langle \pi(g)f_1, \pi(g)f_2\rangle &= \int_{G/H} \langle f_1(g^{-1}g_1), f_2(g^{-1}g_1)\rangle \mathrm{d}\nu_{g_1H} \\ &= \int_{G/H} \langle f_1(g_1), f_2(g_1)\rangle \mathrm{d}\nu_{g_1H}\end{aligned}$$

examples

- $$\mathrm{Ind}_H^G(1) \cong L^2(G/H)$$

Koopman representations

- Let

  $$G \curvearrowright (X, \mathfrak{M}_X, \mu)$$

  be measure preserving group action. Then it induces a **unitary representation**

  $$G \curvearrowright L^2(X)$$

- Let

  $$G \curvearrowright (X, \mathfrak{M}_X, \mu)$$

  be measure *class* preserving group action.
  - Then we define

    $$G \curvearrowright L^2(X)$$
    $$(\pi(g)f)(x) = f(g^{-1}x)\sqrt{\frac{\mathrm{d}g_*\mu}{\mathrm{d}\mu}(x)}$$

    to get a **unitary representation**.
    - To be well-defined we must check the cocycle condition

      $$\pi(g_1g_2)f = \pi(g_1)\pi(g_2)f \iff \frac{\mathrm{d}(g_1g_2)_*\mu}{\mathrm{d}\mu}(x) = \frac{\mathrm{d}g_{1*}\mu}{\mathrm{d}\mu}(x)\frac{\mathrm{d}g_{2*}\mu}{\mathrm{d}\mu}(g_1^{-1}x)$$

      which is true!
  - $$L^2(X, V)$$